Business Context

We operate stationary Battery Energy Storage Systems (BESS) for Commercial & Industrial (C&I) customers in Germany. One of the key use cases is peak shaving - reducing load peaks to lower demand charges, which can represent 30–50% of a C&I customer's electricity bill.

A short-term electricity consumption forecast is a critical input for:

- Charge/discharge scheduling
- Peak shaving threshold decisions
- Operational optimization of the battery

The challenge: Forecast errors have asymmetric costs. Under-forecasting a peak means missing the opportunity to shave it, resulting in real demand charges. Over-forecasting, on the other hand, wastes opportunities to optimize for market prices.

Objective: Build a simple, robust, and interpretable load forecast for a single C&I customer site. This case evaluates:

- Python coding skills
- Data cleaning and preprocessing
- Time-series fundamentals
- Building a pragmatic baseline model
- Interpretation of results in an energy economics context

Time Limit: 3–4 hours: Prioritization is part of the task. We value a complete, working solution over a perfect partial one.

Tasks

Part 1: Data Understanding, Preparation and Exploratory Analysis

Task:

- Explore the provided time series and assess data quality.
- Prepare the data in a way you consider appropriate for further analysis and modeling.
- Analyze load patterns and characteristics relevant for peak shaving.
- Focus on insights that would matter for operational or commercial decisions.

Deliverables:

- Prepared dataset ready for analysis
- Brief documentation of key data issues, assumptions, and preprocessing choices
- A small set of visualizations you consider most informative
- Short summary of the main insights and their relevance for peak shaving

Part 2: Forecasting Approach

Task:

- Design and implement a forecasting approach suitable for supporting peak shaving decisions.

Deliverables:

- An explanation of the modeling choice/approach, validation strategy, and evaluation metric(s)
- Thoughts on the relevance of forecasting horizon and considerations on “what would an appropriate forecasting horizon be

Part 3: Business Interpretation & Reflection

Task:

- Interpret the results from a peak shaving perspective.

Deliverables:

- Discussion of how the forecast could inform peak shaving decisions
- Concise reflection on limitations, risks, and what additional data or improvements would matter most in a production setting

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Deliverables:

- A notebook to follow the exploration and to be used for presentation.
- A clear entry point (e.g. main script or function) so a colleague can easily run the solution.
- Code structured in a way a colleague could read, run, and build on top of without asking questions.

Code Quality:

- We are not looking for production-ready code, but we are looking for code that reflects professional judgment.
- Think of this as an intermediary step. The structure, clarity, and choices you make here tell us more than the model accuracy.